

Sydney Girls High School



MATHEMATICS EXTENSION 1

Year 12

HSC Assessment Task 3

2023

Reading time : 5 minutes
Writing time : 60 minutes
Total marks : 50
Topics: Mathematical Induction; Further Calculus; Motion, Forces and Projectiles

General Instructions:

- There are four (4) questions which **are not** of equal value.
- Attempt all questions.
- Start each question on a NEW page.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- A correct answer without working may be awarded a maximum of 1 mark.
- Write on one side of the paper only.
- Diagrams are NOT to scale, unless indicated otherwise.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.
- A Mathematics reference sheet is provided.

Student Name: _____ **Teacher Name:** _____

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Question 1 (13 marks)

a) Differentiate:

i) $y = \cos^{-1}(7x)$ 1

ii) $y = \tan^{-1}\left(\frac{x}{3}\right)$ 1

b) Find.

i) $\int \frac{1}{16+x^2} dx$ 1

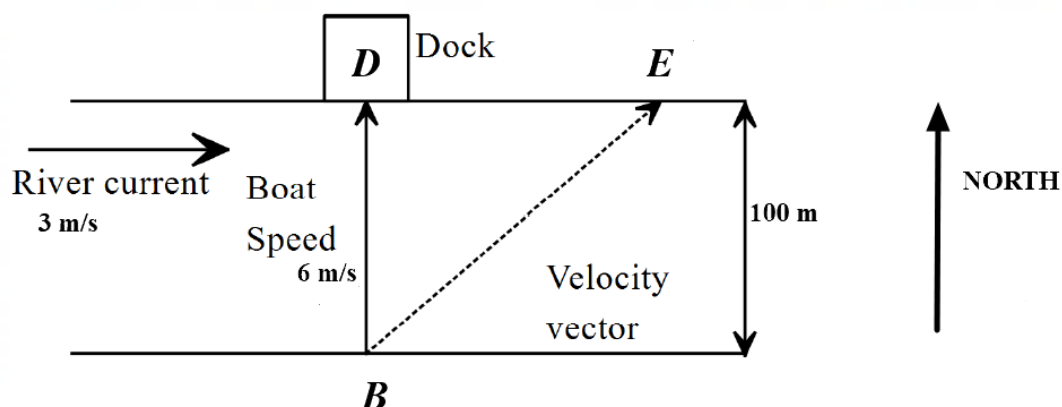
ii) $\int \frac{1}{\sqrt{5-2x^2}} dx$ 2

c) Evaluate the exact value of $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$. 2

d) Find the equation of the tangent to the curve $y = 2 \sin^{-1}(3x) + \frac{\pi}{2}$ at the point where $x = 0$. 2

Question 1 continues on page 4

e)



Breanna has a boat which moves at a top speed of 6 m/s in still water.

From point B , she wants to go due north to point D on the opposite side of the river, as shown in the diagram above.

Today the current in the river is flowing at 3 m/s.

Due to the current, she drifts from point B down the river and arrives at point E .

- i) Taking B as the origin, write down Breanna's velocity vector in the form $x\mathbf{i} + y\mathbf{j}$ and find the magnitude of this vector. 2
- ii) What is the bearing, to the nearest degree, of Breanna's velocity vector? 1
- iii) How far does she travel from B to E ? 1

End of Question 1

Question 2 (12 marks) Begin writing on a NEW page.

- a) Prove by mathematical induction for all integers $n \geq 1$:

3

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n-1)(2n) = \frac{n}{3}(n+1)(4n-1)$$

- b) Use the substitution $u = e^x + 1$ to find $\int \frac{e^x}{\sqrt{e^x + 1}} dx$.

2

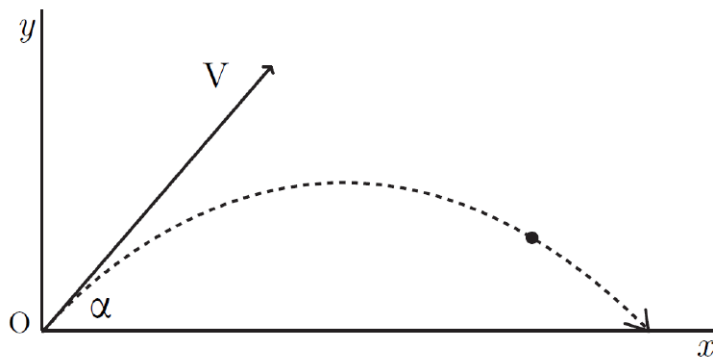
- c) Use the substitution $u = \tan x$ to find $\int \sec^2 x \tan^6 x dx$.

1

- d) A projectile is fired from the origin O with limited speed V in metres per second and with an angle of projection α° . The time is measured in seconds.

The path of the projectile is given by the equations:

$$x = Vt \cos \alpha ; y = Vt \sin \alpha - \frac{1}{2}gt^2, \text{ where } g \text{ is the acceleration due to gravity.}$$



Show all necessary working for the following:

- i) In terms of V and α , find the maximum height h attained by the particle.

2

- ii) Find the range R in terms of V and α .

2

- iii) Show that: $\tan \alpha = \frac{4h}{R}$.

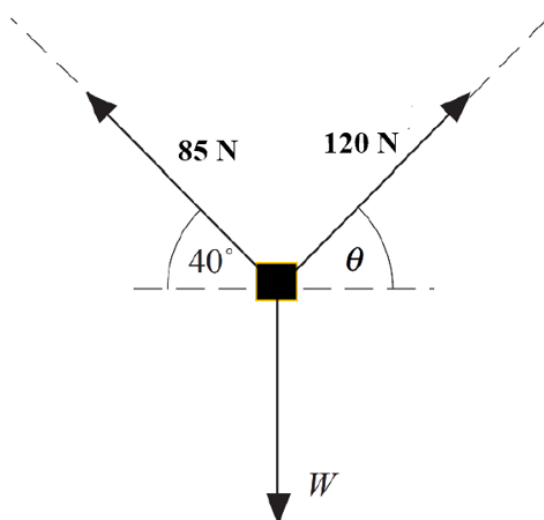
2

End of Question 2

Question 3 (13 marks) Begin writing on a NEW page.

- a) The diagram shows the weight force of gravity, W , that pulls straight down on an object of mass m kg. Two forces of 85 N and 120 N are exerted by cables as shown. The resultant force acting on the object is zero.

- i) Find the angle θ to the nearest degree. 2
- ii) Using your answer from part (i), find the weight force W to the nearest Newton. 2



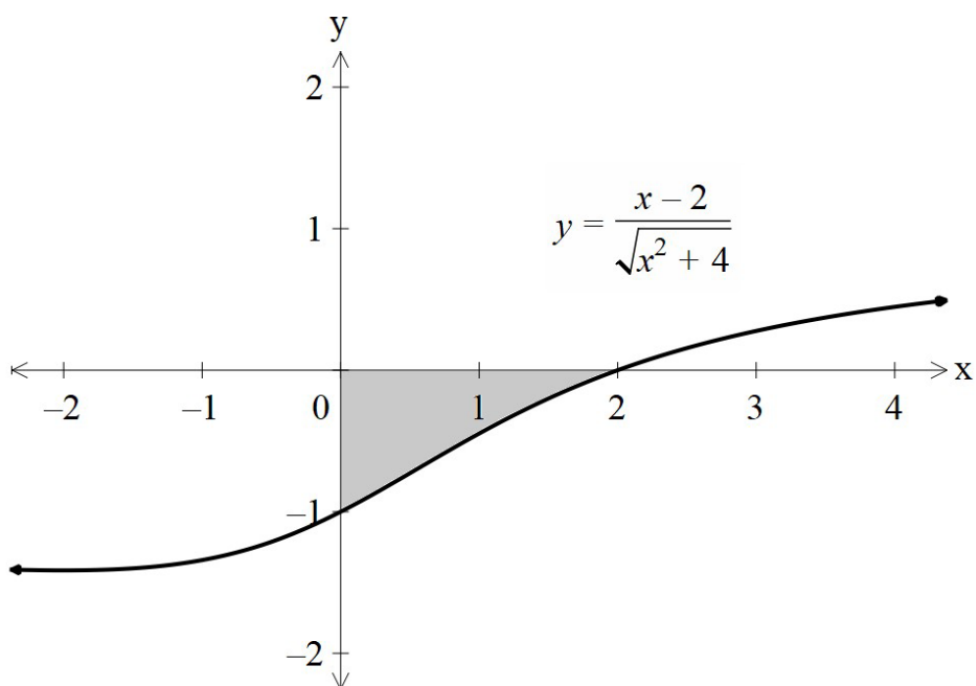
- b)
- i) Show that: $\frac{d}{dx}(x \tan^{-1} x) = \tan^{-1}(x) + \frac{x}{1+x^2}$. Show all working. 1
- ii) Hence, evaluate find the exact value of $\int_0^{\sqrt{3}} \tan^{-1} x \, dx$. 2

Question 3 continues on page 7

c) Prove by mathematical induction that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . 3

d) 3

The shaded region bounded by the curve $y = \frac{x-2}{\sqrt{x^2+4}}$ and $y=0$, from $x=0$ to $x=2$ is shown in the diagram below.



End of Question 3

Question 4 (12 marks) Begin writing on a NEW page.

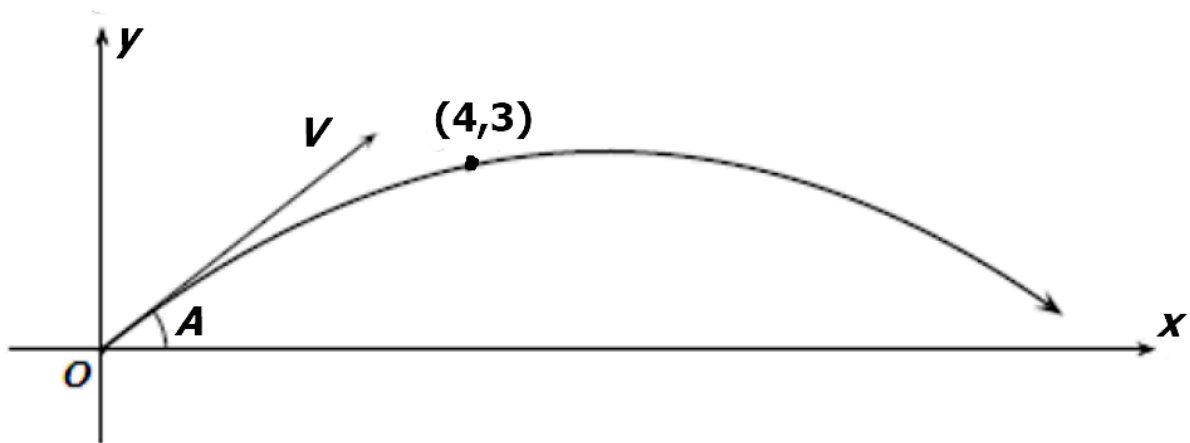
- a) Prove by mathematical induction: $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$
for all integers $n \geq 1$.

3

- b) A particle is projected from a point O with speed of V metres per second at an angle of A to the horizontal. Air resistance is negligible, and the acceleration due to gravity is $g \text{ ms}^{-2}$.

The particle also passes through the point $(4,3)$.

The displacement-time equations for the projectile are: $\underline{r}(t) = Vt \cos A \underline{i} + \left(Vt \sin A - \frac{1}{2}gt^2 \right) \underline{j}$



- i) Show that the Cartesian equation of the trajectory is given by:

$$y = x \tan A - \frac{gx^2}{2V^2} (1 + \tan^2 A).$$

2

Assume $V^2 = 8g$ for part (ii) and part (iii).

- ii) Show that the initial angle A of projection is given by: $A = \tan^{-1}(2)$.

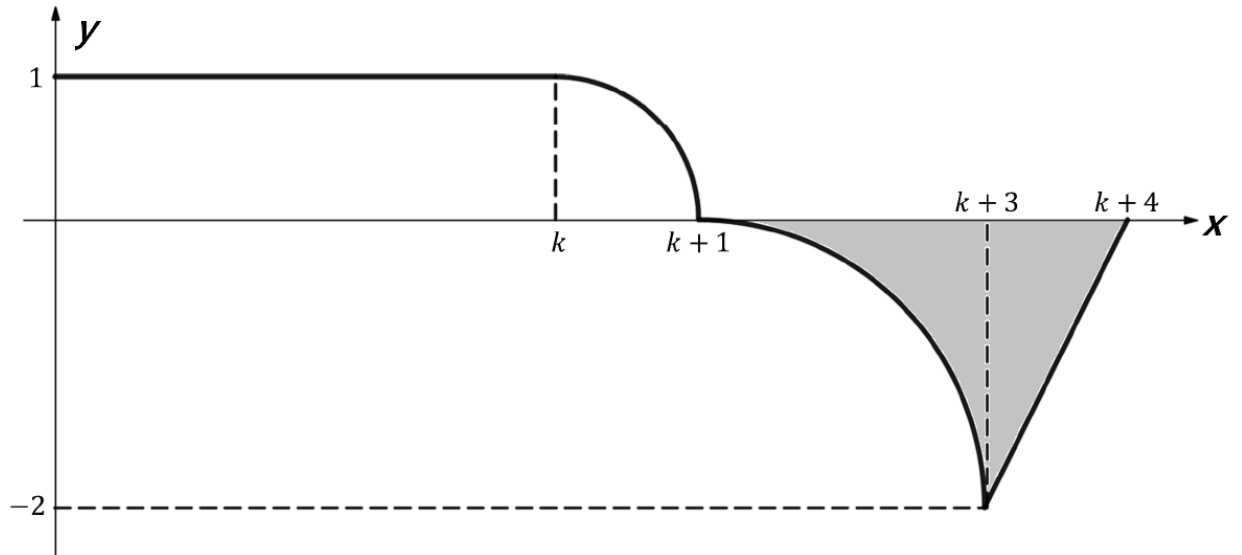
2

- iii) Show that the range R of the projectile is $\frac{32}{5} \text{ m}$.

2

Question 4 continues on page 9

- c) For the given graph it is given that $\int_0^{k+4} f(x) dx = 0$.



- i) Find the value of k in the exact form. 1
- ii) The shaded area is rotated about the x-axis. Find the volume of the solid formed, leave your answer in the exact form. 2

End of Task.

Question 1

a) i) $y = \cos^{-1}(7x)$ ✓

$$y' = \frac{-7}{\sqrt{1-49x^2}}$$

Don't forget to include the correct sign.

ii) $y = \tan^{-1}\left(\frac{x}{3}\right)$

$$y' = \frac{1}{3} \div 1 + \frac{x^2}{9}$$
 ✓

$$= \frac{1}{3} \times \frac{9}{9+x^2}$$

$$= \frac{3}{9+x^2}$$

b) i) $\int \frac{1}{16+x^2} dx = \frac{1}{4} \tan^{-1} \frac{x}{4} + C$ ✓

ii) $\int \frac{1}{\sqrt{5-2x^2}} dx = \frac{1}{\sqrt{2}} \int \frac{\sqrt{2}}{\sqrt{5-2x^2}} dx$ ✓

$$= \frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}x}{\sqrt{5}} + C$$
 ✓

c) $\int_0^{\pi/4} \sin^2 x dx = \int_0^{\pi/4} \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$ ✓

$$= \frac{1}{2} \int (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi}{8} - \frac{1}{4}$$
 ✓

$$d) y = 2 \sin^{-1}(3x) + \frac{\pi}{2}$$

$$y' = 2 \times \frac{3}{\sqrt{1-9x^2}}$$

$$= \frac{6}{\sqrt{1-9x^2}}$$

$$\text{when } x=0 \quad y = 2 \sin^{-1}(0) + \frac{\pi}{2} \quad m_T = 6$$

$$= \frac{\pi}{2}$$

$\therefore$ equation of tangent is:

$$y - \frac{\pi}{2} = 6(x - 0)$$

$$y = 6x + \frac{\pi}{2}$$

$$e) i) \quad \underline{v} = 3\underline{i} + 6\underline{j}$$

Some students mixed up the $\underline{i}$ and $\underline{j}$ components.

$$|\underline{v}| = \sqrt{3^2 + 6^2}$$

$$= \sqrt{45}$$

$$= 3\sqrt{5} \text{ m/s}$$

$$\approx 6.71 \text{ m/s}$$

} both acceptable answers.

$$ii) \quad \tan \theta = \frac{3}{6}$$

$$\tan \theta = \frac{1}{2}$$

$$\theta \approx 27^\circ$$

$\therefore$ Bearing of Breanna's velocity vector is $027^\circ T$.

The answer needed to be given to the nearest degree, as specified in the question, to receive the mark.



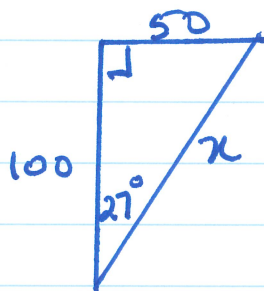
$$\text{iii) } \cos 27 = \frac{100}{x}$$

$$x = \frac{100}{\cos 27}$$

$$x \doteq 112.23 \text{ m}$$



alternate soln:



$$x^2 = 50^2 + 100^2$$

$$x = 50\sqrt{5}$$

$$\doteq 111.8 \text{ m}$$

A range of slightly different answers accepted depending on approach.

Question 2 (12 marks) Begin writing on a NEW page.

a) Prove by mathematical induction for all integers $n \geq 1$:

3

$$1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n-1)(2n) = \frac{n}{3}(n+1)(4n-1)$$

Let $S(n) = (1 \times 2) + (3 \times 4) + (5 \times 6) + \dots + (2n-1)(2n) = \frac{n}{3}(n+1)(4n-1)$

For $n=1$: $LHS = (2n-1)(2n)$ $RHS = \frac{n}{3}(n+1)(4n-1)$
 $= (2-1)(2)$ $= \frac{1}{3}(2)(3)$
 $= 2$ $= 2$

Since $LHS = RHS$, then $S(n)$ is true for $n=1$. ✓

Assume $S(k)$ is true for integer $k \geq 1$.

That is:

$$S(k) = (1 \times 2) + (3 \times 4) + (5 \times 6) + \dots + (2k-1)(2k) = \frac{k}{3}(k+1)(4k-1)$$

Prove that $S(k+1)$ is true.

That is show: $S(k+1) = \frac{(k+1)}{3}(k+1+1)(4(k+1)-1)$

$$S(k+1) = \frac{1}{3}(k+1)(k+2)(4k+3)$$

Proof: $LHS = S(k+1)$

$$= S(k) + T(k+1)$$

$$= \frac{k}{3}(k+1)(4k-1) + [2(k+1)-1][2(k+1)] \quad \left(\begin{array}{l} \text{from} \\ \text{assumption} \end{array} \right)$$

$$= \frac{k}{3}(k+1)(4k-1) + (2k+1)(2k+2)$$

$$= \frac{k}{3}(k+1)(4k-1) + (2k+1)2(k+1)$$

$$= \frac{1}{3}(k+1) [k(4k-1) + 6(2k+1)]$$

$$= \frac{1}{3}(k+1) [4k^2 - k + 12k + 6]$$

$$= \frac{1}{3}(k+1) [4k^2 + 11k + 6]$$

$$= \frac{1}{3}(k+1)(k+2)(4k+3)$$

$$= RHS.$$

Students need to write this at this step.

③

$\therefore s(k+1)$ is true.

Since $s(n)$ is true for $n=1$ and for $n=k+1$ when it is true for $n=k$, then by the Principle of Mathematical Induction, $s(n)$ is true for all integers $n \geq 1$.

* Students who skipped the factorisation step in the proof were deducted one mark.

b) Use the substitution $u = e^x + 1$ to find $\int \frac{e^x}{\sqrt{e^x + 1}} dx$.

2

$$\begin{aligned} u &= e^x + 1 \\ du &= e^x dx \\ \int \frac{e^x dx}{\sqrt{e^x + 1}} &= \int \frac{du}{\sqrt{u}} \\ &= \int u^{-1/2} du \quad \checkmark \\ &= \frac{u^{1/2}}{1/2} + C \\ &= 2\sqrt{u} + C \\ &= \underline{\underline{2\sqrt{e^x + 1} + C}} \quad \checkmark \quad (2) \end{aligned}$$

* Students lost one mark if they did not return their answer in terms of x .

c) Use the substitution $u = \tan x$ to find $\int \sec^2 x \tan^6 x dx$.

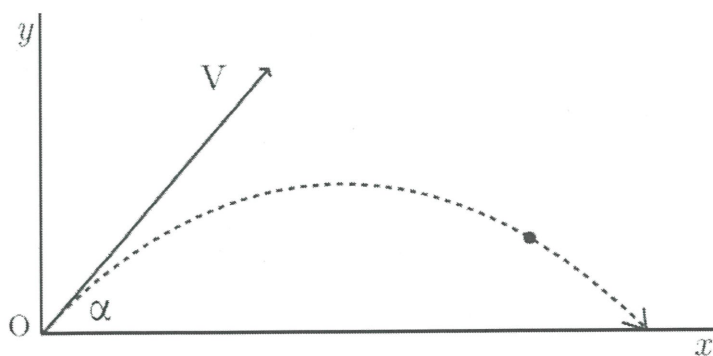
$$\begin{aligned} u &= \tan x \\ du &= \sec^2 x dx \\ \int \sec^2 x \tan^6 x dx &= \int \tan^6 x \sec^2 x dx \\ &= \int u^6 \cdot du \\ &= \frac{u^7}{7} + C \\ &= \frac{\tan^7 x}{7} + C \end{aligned}$$

①
Correct answer
only.

d) A projectile is fired from the origin O with limited speed V in metres per second and with an angle of projection α° . The time is measured in seconds.

The path of the projectile is given by the equations:

$$x = Vt \cos \alpha ; y = Vt \sin \alpha - \frac{1}{2}gt^2, \text{ where } g \text{ is the acceleration due to gravity.}$$



Show all necessary working for the following:

- In terms of V and α , find the maximum height h attained by the particle. 2
- Find the range R in terms of V and α . 2
- Show that: $\tan \alpha = \frac{4h}{R}$. 2

i) Given $y = Vt \sin \alpha - \frac{1}{2}gt^2$

max height occurs when $\dot{y} = 0$:

$$\dot{y} = V \sin \alpha - gt$$

($\dot{y} = 0$) $0 = V \sin \alpha - gt$

$$gt = V \sin \alpha$$

$$t = \frac{V \sin \alpha}{g} \quad \checkmark$$

All formulae
needed to be
derived.

Hence max height, h :

$$\begin{aligned} y &= V \sin \alpha \left(\frac{V \sin \alpha}{g} \right) - \frac{g}{2} \left(\frac{V \sin \alpha}{g} \right)^2 \\ &= \frac{V^2 \sin^2 \alpha}{g} - \frac{V^2 \sin^2 \alpha}{2g} \end{aligned}$$

$$\therefore h = \frac{V^2 \sin^2 \alpha}{2g} \quad \checkmark$$

(2)

ii) Range occurs when $y = 0$:

$$y = Vt \sin \alpha - \frac{gt^2}{2}$$

($y = 0$): $0 = t \left(V \sin \alpha - \frac{gt}{2} \right)$

$$\therefore \text{either } t = 0 \text{ or } V \sin \alpha - \frac{gt}{2} = 0.$$

$$\frac{gt}{2} = V \sin \alpha$$

$$\therefore t = \frac{2V \sin \alpha}{g} \quad \checkmark$$

Hence Range:

$$x = Vt \cos \alpha$$

$$x = V \left(\frac{2V \sin \alpha}{g} \right) \cos \alpha$$

$$\therefore R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

$$R = \frac{V^2 \sin 2\alpha}{g} \quad \checkmark \quad (2).$$

$$\text{iii) } RHS = \frac{4h}{R}$$

$$= \frac{4V^2 \sin^2 \alpha}{2g} \times \frac{g}{2V^2 \sin \alpha \cos \alpha} \quad \checkmark$$

$$= \frac{4V^2 g \sin^2 \alpha}{4V^2 g \sin \alpha \cos \alpha}$$

$$= \frac{\sin \alpha}{\cos \alpha} \quad \checkmark$$

$$= \tan \alpha$$

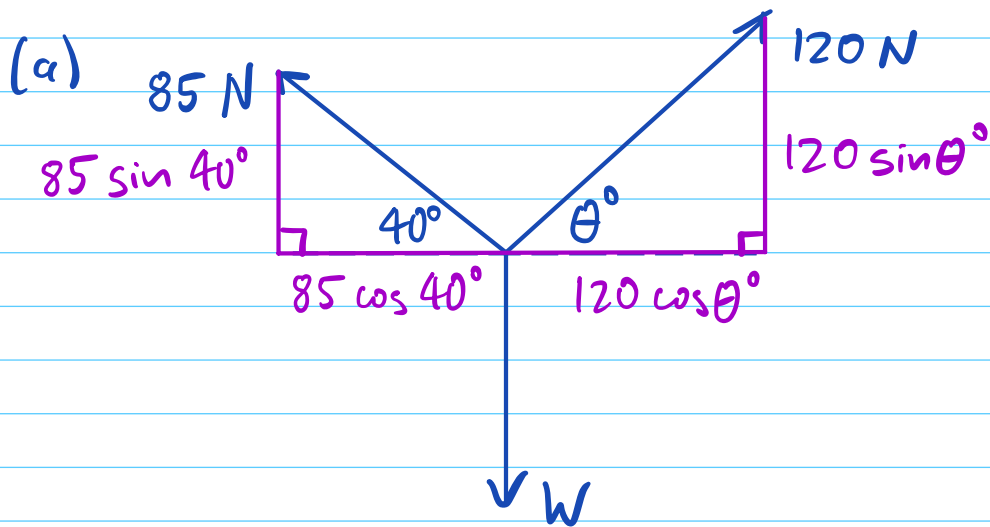
$$= LHS$$

(2)

$$\therefore \tan \alpha = \frac{4h}{R}, \text{ as required.}$$

* Most marks were lost by students in part iii).

Question 3



2

(i) Since the resultant force acting on the object is zero, the net horizontal force is zero. ✓

$$\therefore 120 \cos \theta^\circ - 85 \cos 40^\circ = 0$$

$$\cos \theta = \frac{85 \cos 40^\circ}{120}$$

$$\theta = \cos^{-1} \left(\frac{85 \cos 40^\circ}{120} \right)$$

$$= 57^\circ \text{ (nearest degree)} \quad \checkmark$$

2

(ii) Net vertical force is zero.

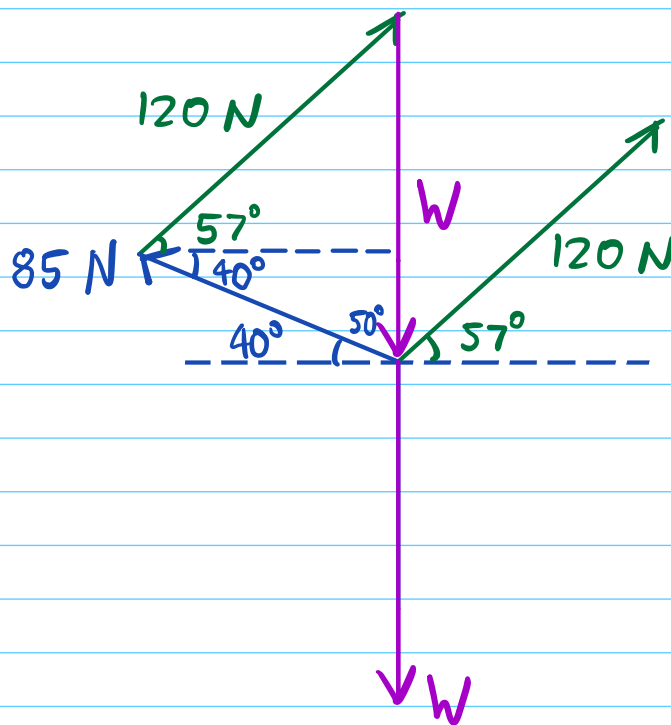
$$\therefore 85 \sin 40^\circ + 120 \sin \theta^\circ - W = 0 \quad \checkmark$$

$$W \doteq 155 \text{ N (nearest newton)} \quad \checkmark$$

Alternative method (Not recommended)

Note: Most students who used this method made at least one mistake in finding angles. The previous method is much safer than this one!

(a) (ii) Take $\theta^\circ \doteq 57^\circ$ from (i).



Note: As the forces are in equilibrium, the vectors can be rearranged to form a triangle, as shown.

$$\frac{W}{\sin 97^\circ} = \frac{120}{\sin 50^\circ} \quad \checkmark \quad (\text{sine rule})$$

$$W = \frac{120 \sin 97^\circ}{\sin 50^\circ}$$

$$\doteq 155 \text{ N} \quad \checkmark \quad (\text{nearest newton})$$

1

$$(b) (i) \frac{d}{dx} \left(\underbrace{x}_u \underbrace{+ \tan^{-1} x}_v \right)$$

$$u = x$$

$$u' = 1$$

$$v = \tan^{-1} x$$

$$v' = \frac{1}{1+x^2}$$

$$= vu' + uv'$$

$$= \tan^{-1} x \times 1 + x \times \frac{1}{1+x^2}$$

$$= \tan^{-1} x + \frac{x}{1+x^2} \checkmark$$

2

$$(ii) \int_0^{\sqrt{3}} \tan^{-1} x \, dx$$

$$= \int_0^{\sqrt{3}} \left(\frac{d}{dx} (x + \tan^{-1} x) - \frac{x}{1+x^2} \right) dx \checkmark$$

$$= \left[x + \tan^{-1} x \right]_0^{\sqrt{3}} - \frac{1}{2} \int_0^{\sqrt{3}} \frac{2x}{1+x^2} dx$$

$$= \sqrt{3} + \tan^{-1} \sqrt{3} - \frac{1}{2} \left[\ln |1+x^2| \right]_0^{\sqrt{3}}$$

$$= \sqrt{3} \left(\frac{\pi}{3} \right) - \frac{1}{2} \ln 4 \checkmark$$

$$= \frac{\pi\sqrt{3}}{3} - \ln 2$$

3

(c) Base case: For $n=1$,

$$9^{n+2} - 4^n = 9^{1+2} - 4^1$$

$$= 729 - 4$$

$$= 725$$

$$= 5 \times 145, \quad \checkmark$$

which is divisible by 5.

Assume true for $n=k$:

$$9^{k+2} - 4^k = 5P,$$

where P is a positive integer.

$$\text{i.e.} \quad 9^{k+2} = 5P + 4^k \quad (*) \quad \checkmark$$

Prove true for $n=k+1$:

$$9^{k+3} - 4^{k+1} = 9(\underline{9^{k+2}}) - 4(4^k)$$

$$\text{From } (*) \rightarrow = 9(\underline{5P + 4^k}) - 4(4^k)$$

$$= 5(9P) + 9(4^k) - 4(4^k)$$

$$= 5(9P) + 5(4^k)$$

$$= 5(9P + 4^k)$$

which is divisible by 5, as required. $\checkmark$

3

$$(d) \text{ Volume} = \pi \int_0^2 y^2 dx \quad \checkmark$$

$$= \pi \int_0^2 \left(\frac{x-2}{\sqrt{x^2+4}} \right)^2 dx$$

$$= \pi \int_0^2 \frac{(x-2)^2}{x^2+4} dx$$

$$= \pi \int_0^2 \frac{x^2 - 4x + 4}{x^2 + 4} dx$$

$$= \pi \int_0^2 \left(\frac{x^2 + 4}{x^2 + 4} - \frac{4x}{x^2 + 4} \right) dx \quad \checkmark$$

$$= \pi \int_0^2 \left(1 - \frac{4x}{x^2 + 4} \right) dx$$

$$= \pi \left[x - 2 \ln |x^2 + 4| \right]_0^2$$

$$= \pi [2 - 2 \ln 8 - (0 - 2 \ln 4)] \quad \checkmark$$

$$= \pi(2 - 2\ln 8 + 2\ln 4)$$

$$= \pi(2 - 6\ln 2 + 4\ln 2)$$

$$= 2\pi(1 - \ln 2) \text{ cubic units}$$

Notes:

(c) The best method for the induction question was to sub $q^{k+2} = 5M + 4^k$ into $q^{k+3} = 9(q^{k+2})$. Students who tried to use manipulation often made a mistake with $\pm$ signs.

(d) No need to use substitution for $\int_0^2 \frac{4x}{x^2+4} dx$
 Students are encouraged to recognise that the top is almost the same as the derivative of the bottom, which leads to $\left[2 \ln |x^2+4| \right]_0^2$.

Q4

$$a) 1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

prove it is true for $n = 1$

$$LHS = 1 \times 1! = 1$$

$$RHS = (1+1)! - 1 = 1$$

LHS = RHS

True for $n = 1$

Assume it is true for $n = k$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! = (k+1)! - 1$$

prove it is true for $n = k+1$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! + (k+1)(k+1)! = (k+2)! - 1$$

$$LHS = (k+1)! - 1 + (k+1)(k+1)!$$

$$= (k+1)! [k+1 + 1] - 1$$

$$= (k+2)(k+1)! - 1$$

$$= (k+2)! - 1$$

$$= RHS. \text{ This is true for } n = k+1$$

It is proven by Mathematical Induction.

Q4

$$b) \quad x = Vt \cos A \quad (1)$$

$$y = -\frac{1}{2}gt^2 + Vt \sin A \quad (2)$$

i) from (1) $t = \frac{x}{V \cos A}$ subs into (2)

$$y = -\frac{1}{2}g \left(\frac{x^2}{V^2 \cos^2 A} \right) + V \left(\frac{x}{V \cos A} \right) \sin A$$

$$y = x \tan A - \frac{gx^2}{2V^2} \sec^2 A$$

$$y = x \tan A - \frac{gx^2}{2V^2} (1 + \tan^2 A) \quad (*)$$

ii) use $V^2 = 8g$ and $x = 4$, $y = 3$
subs into (*)

$$3 = 4 \tan A - \frac{g \times 16}{2 \times 8g} (1 + \tan^2 A)$$

$$3 = 4 \tan A - 1 - \tan^2 A$$

$$\tan^2 A - 4 \tan A + 4 = 0$$

$$(\tan A - 2)^2 = 0$$

$$\tan A = 2$$

$$A = 63^\circ 26'$$

Q4/b

iii) from (i)

$$y = x \tan A - \frac{gx^2}{2v^2} (1 + \tan^2 A)$$

$$\text{Subs } \tan A = 2, \quad v^2 = 8g, \quad y = 0$$

Range = Max horizontal disp $\therefore y = 0$

$$0 = 2x - \frac{gx^2}{16g} (1 + 4)$$

$$2x - \frac{5x^2}{16} = 0$$

$$32x - 5x^2 = 0$$

$$x(32 - 5x) = 0$$

$$x = 0$$

$$x = \frac{32}{5}$$

$$\therefore \text{Range} = \frac{32}{5} = 6.4 \text{ m}$$

$$\boxed{Q4} \text{ c/i) } \int_0^{k+4} f(x) dx = 0 \text{ means}$$

top Areas = bottom Areas

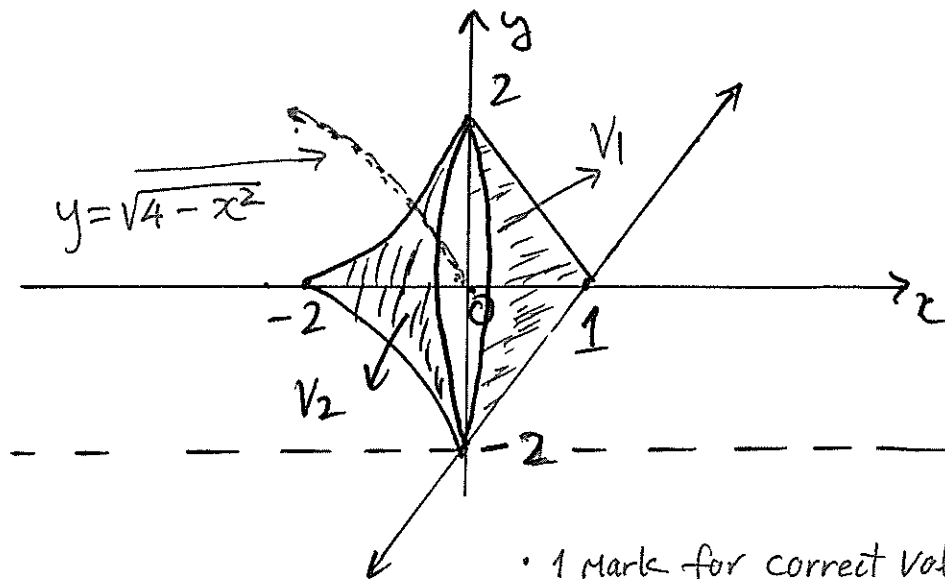
$$\text{Top Areas} = (k \times 1) + \frac{\pi \times 1^2}{4} = k + \frac{\pi}{4}$$

$$\begin{aligned} \text{Bottom Areas} &= \left(2 \times 2 - \frac{\pi \times 2^2}{4}\right) + \left(\frac{1 \times 2}{2}\right) \\ &= 5 - \pi \end{aligned}$$

$$k + \frac{\pi}{4} = 5 - \pi$$

$$k = 5 - \frac{5\pi}{4}$$

[Q4] c/ii) The diagram can be drawn as



• 1 Mark for correct Vol of the Cone.
OR correct Vol of V_2
OR correct Area under the x -axis.

$$V = V_1 + V_2$$

V_1 = Vol of a cone, radius = 2 u, height = 1 u

V_2 = Vol of the curve $\sqrt{4-x^2} - 2$ rotated about the x -axis from -2 to 0.

$$V_1 = \frac{\pi r^2 h}{3} = \frac{\pi \times 2^2 \times 1}{3} = \frac{4\pi}{3}$$

$$V_2 = \pi \int_{-2}^0 y^2 dx = \pi \int_{-2}^0 (\sqrt{4-x^2} - 2)^2 dx$$

$$\begin{aligned} V_2 &= \pi \int_{-2}^0 (4 - x^2 - 4\sqrt{4-x^2} + 4) dx \\ &= \pi \int_{-2}^0 (8 - x^2) dx - \pi \int_{-2}^0 4\sqrt{4-x^2} dx \\ &= \pi \left[8x - \frac{x^3}{3} \right]_{-2}^0 - 4\pi \left[\frac{1}{4} \pi \times 2^2 \right] = \frac{40\pi}{3} - 4\pi^2 \end{aligned}$$

$$\text{Total Vol: } V = V_1 + V_2 = \frac{4\pi}{3} + \frac{40\pi}{3} - 4\pi^2 = \frac{44\pi}{3} - 4\pi^2$$